

Class 8: A Brief History of Wi-Fi

Protocols, PHY Waveforms, and the Design Choices from
802.11b to 802.11be

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CIS 4930/5930 Future Edge Networks
xinliulab.github.io/FSU-CIS4930-CIS5930-Future-Edge-Networks

Wi-Fi grew by spending new resources



The icon never changed. The radio under it did — six ways.

Preview, not prerequisite — every row gets its own slide today. Keep a tally.

Dimension	How Wi-Fi spent it
Spectrum	2.4 → 5 → 6 → 60 GHz

Arrows list resources, not chronology: 60 GHz (2012) predates 6 GHz (2020s).

One family identity; each amendment spends a new dimension.

QAM = Quadrature Amplitude Modulation MIMO = Multiple Input, Multiple Output
OFDMA = Orthogonal Frequency-Division Multiple Access

IEEE 802.11 standard family; IEEE 802.11be; Cisco 802.11ax white paper



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Links	one band → multi-link operation

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Amendment names vs. Wi-Fi generations

IEEE amendment	Market name	PHY shorthand
802.11b	Wi-Fi 1	HR-DSSS / CCK
802.11a	Wi-Fi 2	OFDM at 5 GHz
802.11g	Wi-Fi 3	OFDM at 2.4 GHz
802.11n	Wi-Fi 4	HT: MIMO + 40 MHz
802.11ac	Wi-Fi 5	VHT: wider + 256-QAM
802.11ax	Wi-Fi 6 / 6E	HE: OFDMA + 1024-QAM
802.11be	Wi-Fi 7	EHT: MLO + 320 MHz
802.11ad / ay	WiGig	directional 60 GHz

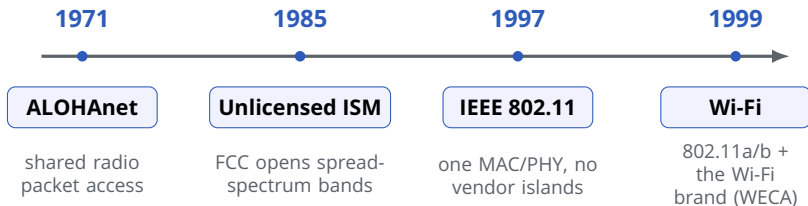
IEEE names amendments; the Wi-Fi Alliance names generations (Wi-Fi 1–3 are retroactive, informal). The orange ad/ay row is a parallel branch, outside Wi-Fi 1–7. Third column = today's itinerary — every acronym gets its own slide. Read columns 1–2 now.

One radio, two naming systems: engineering and marketing.

HR = High Rate HT/VHT/HE/EHT = High / Very High / High-Efficiency / Extremely High Throughput MLO = Multi-Link Operation

[IEEE 802.11 interactive timeline; Keysight 802.11ad concepts](#)

Four milestones made Wi-Fi possible



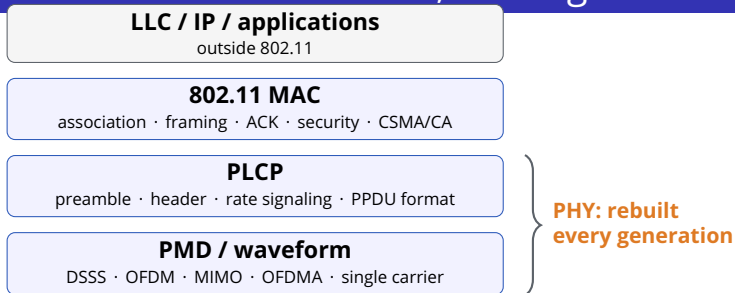
Q: Which of the four is regulatory rather than technical?

A: 1985. The FCC opened the “junk bands” microwave ovens pollute — nobody needed permission to ship a radio there, and Wi-Fi built an empire on that permissionless ground.

ISM = Industrial, Scientific, and Medical bands FCC = Federal Communications Commission

IEEE 802.11 interactive timeline; Wi-Fi Alliance (WECA) newsroom, first certifications April 2000

The 802.11 stack: stable MAC, moving PHY



PPDU = the PHY-layer transmission unit placed on air: preamble + header + payload.

Same family behavior above, different ways to turn bits into radio energy below. That split is Wi-Fi's longevity.

MAC = Medium Access Control PLCP = Physical Layer Convergence Procedure PMD = Physical Medium Dependent association = joining an AP's network (1997 split; later revisions fold PLCP/PMD into one PHY)

IEEE Std 802.11 (MAC and PHY specifications)

1997 solved shared access before speed

Ethernet detects collisions mid-send. Why can't a radio?



DIFS (Distributed Inter-Frame Space) = a fixed polite pause after the channel goes quiet

Original PHY choices

- 1 or 2 Mb/s
- FHSS or DSSS at 2.4 GHz
- Infrared (never shipped)

Because a radio...

- Own TX deafens own RX
- Not every station is audible
- So: back off, then demand ACK

CSMA/CA avoids collisions; it cannot detect them while transmitting. Silence proves nothing — only an ACK does.

CSMA/CA = Carrier Sense Multiple Access / Collision Avoidance FHSS/DSSS = Frequency-Hopping / Direct-Sequence Spread Spectrum

IEEE Std 802.11-1997 baseline

802.11b: coded spreading at 2.4 GHz

11 Mb/s max PHY rate $\approx$ **22 MHz** channel **2.4 GHz** band

one symbol

→



a patterned chip sequence (conceptual; Barker = 11 chips, CCK = 8 complex chips)

- **1 / 2 Mb/s:** Barker spreading, as in 1997
- **5.5 / 11 Mb/s:** CCK — the chosen pattern is the data
- Receiver knows the pattern — it digs the signal out of noise

$\approx$ 22 MHz-wide channels on 5 MHz centers is why
North America gets only three clean 2.4 GHz channels.

DSSS = Direct-Sequence Spread Spectrum CCK = Complementary Code Keying — bits ride in the code words' phases chip = one element of the pattern processing gain = the technical name for that noise-digging

[IEEE 802.11 interactive timeline; IEEE Std 802.11](#)

802.11a threw the single carrier away

802.11b took one carrier and spread it with a cleverer pattern. That road has an end: the faster you push one wave, the more a wall echo smears one symbol into the next.

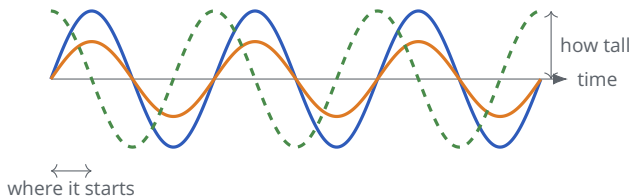
802.11a answers with OFDM (Orthogonal Frequency Division Multiplexing) — not one fast wave, but 64 slow ones at once.

You have heard the name; almost nobody has seen the inside. We build it from the ground up in four parts — then 802.11a is one slide:

- 1 What a radio actually sends — and why that is a point.
- 2 Why many waves at once, and why they must be perpendicular.
- 3 The FFT — the machine that adds waves up and takes them apart.
- 4 Transmitter, receiver, and the grid.

Everything through 802.11b runs on what these four parts build.

A radio sends a wave, not bits

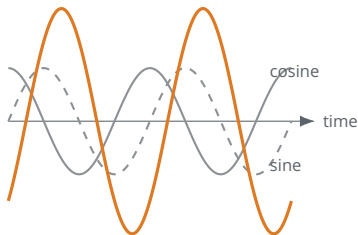


blue: tall, starts at 0 **orange:** half as tall **green:** tall, starts a quarter cycle early

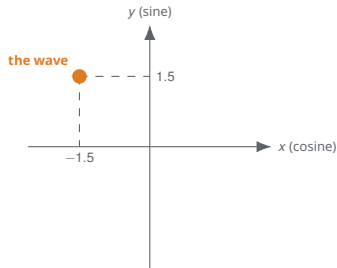
An antenna cannot emit a 0 or a 1. It emits a **wave** at some frequency. At one frequency a wave has exactly two things you can change: **how tall it is** (amplitude) and **where in its cycle it starts** (phase). Everything else about it is fixed by the frequency.

Two knobs. To send information, we turn the knobs.

Two knobs are one point



$$\text{orange} = -1.5 \times \text{cosine} + 1.5 \times \text{sine}$$



Any wave of the chosen frequency — any height, any starting point — can be written as “x parts of a cosine plus y parts of a sine”. So the two knobs are just two numbers, and two numbers are a **point** (x, y).

One point = one wave. From now on, when you see a dot, think “a wave”.

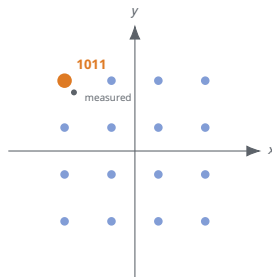
Bits pick the point

Agree on 16 allowed points — 16 allowed waves (the scheme is called 16-QAM; the name does not matter).
Cut the bits into groups of four:

1011 0010 1110 0001 ...

Each group names one point. The radio sends that wave for one **symbol time** T , then the next group's wave, and so on.

The receiver measures (x, y) , picks the **nearest** allowed point, and reads four bits back. Noise nudges the measurement, so allowed points must keep their distance — hence 16, not thousands.



16 allowed waves, 4 bits each

One wave carries 4 bits every T : slow.

Part 2: one wave is slow — so use many

Four bits per symbol time is nowhere near a video stream. Two ways to go faster:

Option A: one wave, shorter symbols.

Keep one wave and switch points 64 times faster. Symbol time shrinks from T to $T/64$.

Option B: 64 waves at once.

Use 64 different frequencies at the same time, each carrying its own point for the full time T . Add them up and send the sum.

Both give 64 times the bits. Every modern radio — Wi-Fi, 4G, 5G — chose **B**. The next slide is the reason, and it is the same physics this whole lecture is about.

Same speed. So why does everyone pick many slow waves over one fast one?

Why not one fast wave? Echoes.

A: short symbols



echo lands on the next symbol

B: long symbols



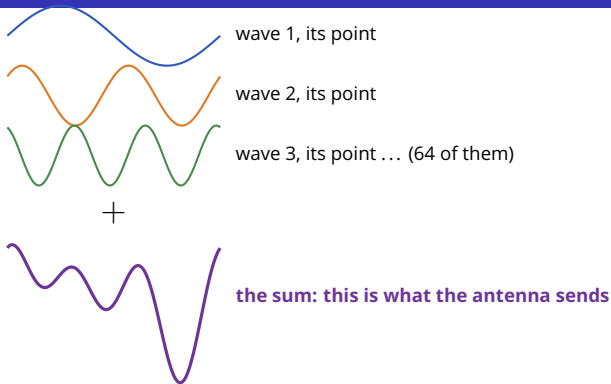
echo mostly lands on itself

↔
echo delay

Radio bounces. A copy of your signal arrives a little late off a wall or a building — about $1\ \mu\text{s}$ later for every 300 m of extra path. With short symbols (A) that echo is a smeared copy of the previous symbol on top of the current one: garbage. With long symbols (B) the echo overlaps the same symbol almost entirely; it just makes the wave a bit taller or shorter, which the receiver can undo.

Echoes are why we send many slow waves instead of one fast one.

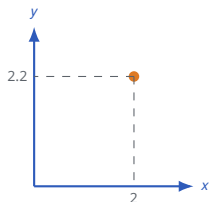
Add 64 waves and send the sum — then what?



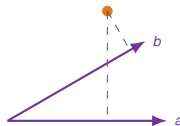
The receiver gets only the purple line. It has to recover **each** wave's (x, y) from it — 64 pairs of numbers out of one squiggle.
Looking at the squiggle, that seems hopeless.

When can you pull one wave back out of a sum of many?
That is the question the word “orthogonal” answers.

Perpendicular axes: reading one ignores the other



perpendicular: reading x
picks up nothing of y



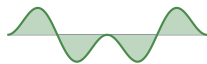
tilted: reading a
picks up part of b

Think of a point on a plane. With perpendicular axes, its x -reading does not change when you slide it along y : the two numbers are independent, each can be read on its own. Tilt the axes and that breaks — moving along b changes the a -reading. The two numbers leak into each other.

Separable $\Leftrightarrow$ perpendicular. Now we need the same idea for waves.

Perpendicular waves, drawn

1 cycle $\times$ 2 cycles in time T



product

area above = area below: total 0

1 cycle $\times$ 1.5 cycles in time T



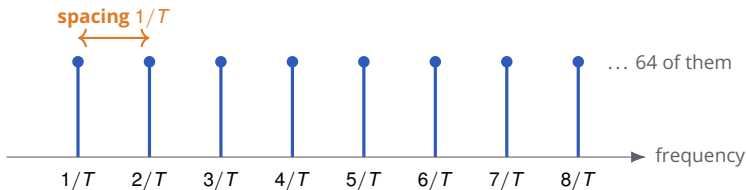
product

more above than below: total $\neq 0$

To measure “how much of wave 1 is in the sum”, the receiver multiplies the sum by wave 1 and adds up over the symbol time T . Two waves are **perpendicular** when that product adds up to **zero**: measuring one picks up nothing of the other. Left: it works. Right: it does not — wave 2 leaks into the reading.

Whole cycles in T : perpendicular. Fractional cycles: leak.

The perpendicular family: frequencies k/T

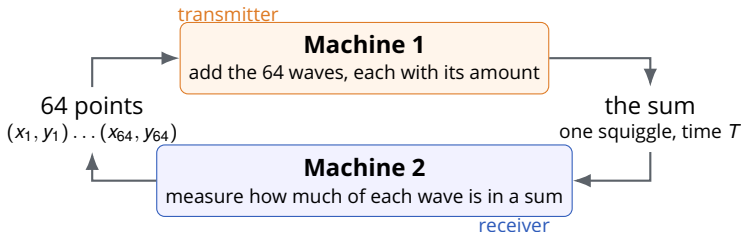


A wave fits a whole number of cycles into T exactly when its frequency is a multiple of $1/T$. So the waves at $1/T, 2/T, 3/T, \dots$ are **all perpendicular to each other**. That is the family OFDM uses. Each member is called a **subcarrier**, and their spacing is $1/T$ — not a design choice, a consequence.

Real numbers. Wi-Fi: $T = 3.2 \mu\text{s}$, so subcarriers 312.5 kHz apart — remember that pair, the next slides keep using it. 5G: $T \approx 67 \mu\text{s}$, spacing 15 kHz (30, 60, 120 kHz variants exist).

We now know which 64 waves to use.

Part 3: we need two machines



Both machines work on the same 64 waves. To put them in a computer, sample each wave at 64 instants during T : a wave becomes a row of 64 numbers, and 64 waves become a 64×64 table. **Machine 2 is that table.** Machine 1 is the same table run backwards.

That table has a name: the FFT matrix.

Two waves: the 2×2 FFT, with numbers

Machine 2 — measure. Two waves sampled at two instants: a flat one $(1, 1)$ and a flipping one $(1, -1)$. Stack them as rows:

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$$

The sum $(3, 1)$ contains 4 units of the flat wave and 2 of the flipping one. Rows perpendicular:
 $1 \cdot 1 + 1 \cdot (-1) = 0$.

Machine 1 — add. Same table, run backwards (divide by 2):

$$\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

Give it the amounts $(4, 2)$ and it rebuilds the sum $(3, 1)$. Round trip complete, nothing lost — because the rows are perpendicular.

FFT = measure each wave. IFFT = add the waves back.
Perpendicular rows are what make the round trip exact.

Four waves: the 4×4 FFT

Four waves at $0, 1/T, 2/T, 3/T$, each sampled at four instants:

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{pmatrix}$$

Row k is the wave at frequency k/T . The $j (= \sqrt{-1})$ is only there so a wave can rotate through its cycle instead of merely flipping sign.

Check one pair. Row 2 against row 4 (flip the sign of j in one of them, multiply entry by entry, add):

$$\begin{aligned} 1 \cdot 1 + (-j)(-j) + (-1)(-1) + j \cdot j \\ = 1 - 1 + 1 - 1 = 0 \end{aligned}$$

Zero. Try any other pair — also zero. **Every wave is perpendicular to every other**, exactly as the drawing promised.

The 64×64 table is the same thing, bigger. The 4096×4096 one too.

The FFT matrix is the list of perpendicular waves, written as numbers.

Why “Fast” is the important letter

Multiplying a signal by an $N \times N$ table costs N^2 multiplications. The FFT gets the identical answer in about $N \log_2 N$ — by reusing the halves of the table, which are copies of each other.

waves N	plain table N^2	FFT $N \log_2 N$	saving
64	4 096	384	11×
1 024	1 048 576	10 240	100×
4 096	16 777 216	49 152	340×

A Wi-Fi chip runs this for every symbol, tens of thousands of times a second, on a battery. With the plain table it could not. The FFT algorithm (Cooley–Tukey) dates from 1965; the first paper proposing OFDM built on it is from 1971. Not a coincidence — and 802.11a, three slides from now, picks $N = 64$: the top row.

OFDM is practical because the FFT is fast.

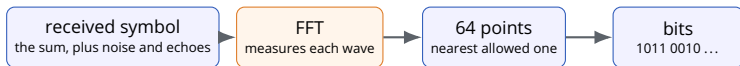
Part 4: the transmitter



- Take 256 bits. Cut into 64 groups of four; each group picks a point.
- Put point k on the wave at frequency k/T — on **subcarrier** k .
- One IFFT adds the 64 waves into one burst of length T : the **symbol**.
- Send it. Then the next 256 bits, the next symbol, and so on.

256 bits in, one symbol out — 64 times what one wave could do.

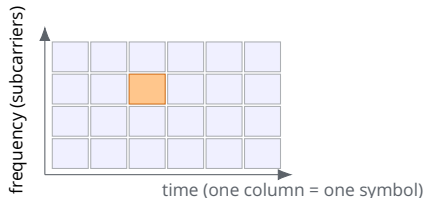
The receiver



- One FFT on the received burst: out come 64 (x, y) pairs, one per subcarrier — cleanly, because the waves are perpendicular.
- Each pair is nudged by noise; snap it to the nearest allowed point; read 4 bits.
- The echo from slide “Why not one fast wave” is in here too. It makes each subcarrier’s point a little larger or rotated — a per-subcarrier correction, then forgotten. Or kept: that echo is what the radar half of ISAC reads.

Same 64 waves, same FFT, read the other way round.

Symbol after symbol: the grid



- Each column is one symbol; each row is one subcarrier.
- The orange cell: subcarrier 3 during symbol 3 — **one wave, one symbol time, one point, four bits.**
- 5G calls a cell a **resource element**. Wi-Fi 6 hands out blocks of them called **resource units** — that is OFDMA, later today.

Every OFDM radio — Wi-Fi, 4G, 5G — fills and reads this grid, symbol after symbol.

802.11a: many orthogonal subcarriers

54 Mb/s max rate **20 MHz** channel **5 GHz** band



at each dashed sampling line, every neighbor is exactly zero (conceptual)

Nothing new on this slide: this is the family k/T we just built, with $T = 3.2 \mu\text{s}$ and 64 of them across 20 MHz. The comb is the picture; the FFT matrix is the same thing as numbers.

64-point FFT · 48 data + 4 pilot subcarriers · BPSK-64-QAM · 6-54 Mb/s

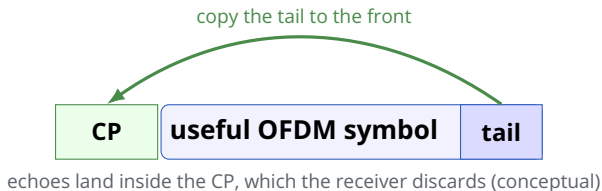
Q: If OFDM was better, why did 802.11b win the first round?

A: 11a silicon came ~ 2 years late (Atheros, late 2001), cost more, crossed walls worse, and couldn't talk to 11b. A great waveform loses to a cheap, entrenched ecosystem — for a while.

OFDM = Orthogonal Frequency-Division Multiplexing FFT = Fast Fourier Transform pilots = known reference tones that keep the receiver calibrated

[IEEE Std 802.11 \(OFDM PHY\); IEEE 802.11 timeline](#)

Why OFDM needs a cyclic prefix



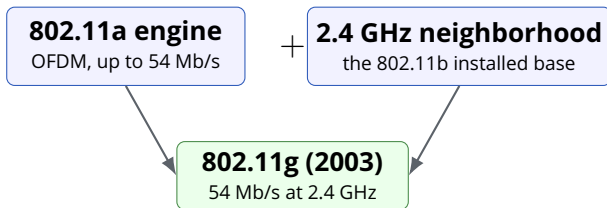
- Delay spread becomes a bounded guard region
- Each subcarrier sees one complex gain
- Equalization = one multiply per subcarrier

The cyclic prefix turns multipath smearing into a fixed, budgeted overhead — paid by every OFDM Wi-Fi since 11a.

CP = Cyclic Prefix — Wi-Fi calls this the guard interval (GI) delay spread = arrival-time spread of echoes equalization = undoing what the channel did; here, one multiply per subcarrier complex gain = the channel's single scale-and-shift per tone

IEEE Std 802.11 (OFDM PHY, guard interval)

802.11g: the OFDM engine comes home



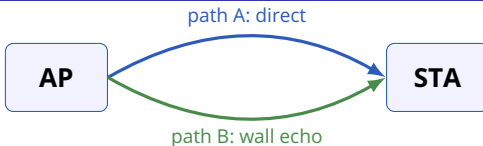
- Same OFDM waveform, more popular band
- Legacy 11b stations cannot decode OFDM
- So mixed networks send protection frames first

Backward compatibility bought adoption and cost airtime: protection overhead is the rent paid to the installed base.

protection frame = a short legacy-format announcement that reserves the channel before OFDM speaks

[IEEE 802.11 interactive timeline; IEEE Std 802.11](#)

802.11n: multipath becomes an asset



more independent paths → more solvable equations → more streams (conceptual)

stream = an extra simultaneous data lane through the same channel and band.

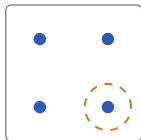
- **40 MHz** channel bonding
- Up to **4 SU-MIMO** streams
- Short guard interval
- **A-MPDU** frame aggregation

600 Mb/s ceiling = 4 streams × 40 MHz × 64-QAM 5/6
× short GI. Typical clients implemented far fewer streams.

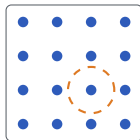
SU-MIMO = Single-User MIMO A-MPDU = Aggregated MAC Protocol Data Unit STA = station (client) 5/6 = coding rate — the data fraction; the rest is error protection

[IEEE 802.11 interactive timeline; IEEE Std 802.11n-2009](#)

What “tighter modulation” means



4-QAM: 2 bits



16-QAM: 4 bits

same noise blob: harmless when points are far apart,
fatal when they are dense (conceptual)

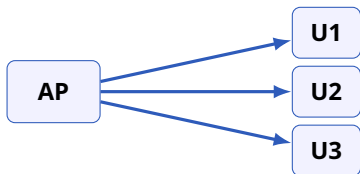
A **constellation** = the set of allowed signal points.
More bits per symbol = more points, packed closer — so noise flips them.

Denser needs cleaner: every step up the QAM ladder
(64 → 256 → 1024 → 4096) demands a higher SNR.

SNR = Signal-to-Noise Ratio — how clean the signal is QAM points differ in amplitude and phase
noise blob = where the received point may land

802.11ac: wider, tighter, more users

- **Wider:** up to 160 (80+80) MHz
- **Tighter:** 256-QAM (8 bits)
- **More streams:** 8 maximum
- **DL MU-MIMO:** Wave 2 gear



downlink MU-MIMO: one transmission, several spatially separated users

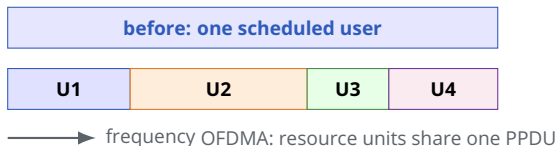
Q: Will any laptop ever see the 6.9 Gb/s on the box?

A: No. A maximum PHY rate is a configuration ceiling — every optimistic option at once — not expected application throughput.

MU-MIMO = Multi-User MIMO DL = downlink “Wave 2” = a product-certification wave, not a standards phase

Cisco, Migrating to 802.11ac; Cisco 802.11ac throughput testing note

802.11ax: dense efficiency, not peak



- **OFDMA:** frequency RUs
- **UL + DL MU-MIMO**
- **1024-QAM:** 10 bits
- **BSS coloring:** ignore faint cells
- **TWT:** the AP schedules naps

No contradiction with CSMA/CA: “faint” = too weak to collide at my receiver — coloring skips a needless deferral.

New target: dense efficiency, not a bigger single-link headline.

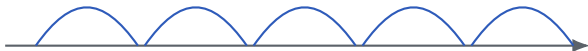
HE = High Efficiency RU = Resource Unit BSS = Basic Service Set cell = one AP's coverage (a BSS) TWT = Target Wake Time UL = uplink coloring = a per-cell tag on frames

Cisco 802.11ax white paper; IEEE Std 802.11

HE OFDM: four times narrower subcarriers

802.11a/g/n/ac

312.5 kHz spacing · 3.2 μ s useful symbol



802.11ax HE

78.125 kHz spacing · 12.8 μ s useful symbol



same 20 MHz channel, 4 $\times$ more FFT points (conceptual, not a mask)

- Longer symbol $\rightarrow$ guard interval is relatively cheaper
- More subcarriers $\rightarrow$ finer-grained resource units
- Guard interval adapts: 0.8 / 1.6 / 3.2 μ s options

Narrower subcarriers are what make small OFDMA resource units — and outdoor-tolerant symbols — possible at all.

[Cisco 802.11ax white paper \(HE OFDM numerology\)](#)

Wi-Fi 6E adds spectrum, not a waveform

2.4 GHz

range + legacy

5 GHz

capacity, mature

6 GHz

clean wide channels

same HE PHY/MAC toolbox — new operating spectrum and channel plan

- 6E = 802.11ax certified into 6 GHz
- No legacy stations → no rent to the installed base
- Channel availability varies by country

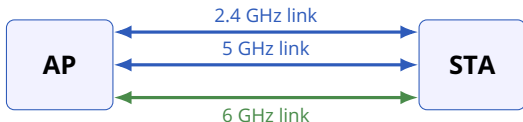
Q: Is Wi-Fi 6E a new generation?

A: No. A band extension is not a new IEEE amendment. The waveform is Wi-Fi 6's; the gift is clean, wide, uncontested spectrum.

[IEEE Std 802.11; Cisco Wi-Fi 6/6E materials](#)

802.11be: several links act as one

Problem: one association = one band; a busy band jails every retry.



multi-link operation: one association, several coordinated radios

- **320 MHz** (6 GHz only)
- **4096-QAM:** 12 bits, high SNR
- **≥ 30 Gb/s** required mode
- **Puncturing:** skip busy MHz

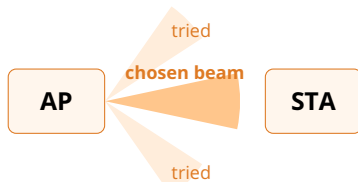
MLO spends the last resource dimension: links themselves — throughput, balancing, redundancy, or latency, per traffic.

EHT = Extremely High Throughput MAC data SAP = MAC data service access point, where ≥ 30 Gb/s is measured puncturing = preamble puncturing (the IEEE term), unrelated to error-code puncturing

IEEE Std 802.11be-2024

802.11ad: enormous bandwidth at 60 GHz

2.16 GHz channel **4.62 Gb/s** SC max rate **room-scale** range



severe path loss → the link closes only with aimed antenna gain
beam training: sweep sectors, keep the winner (conceptual)

- **Control PHY:** robust spread BPSK
- **Single-carrier PHY:** 4.62 Gb/s workhorse
- **OFDM PHY:** ≈ 6.8 Gb/s option, later obsoleted

Rate from raw bandwidth, not dense modulation. The price:
walls end the link, and every link must first be aimed.

Keysight 802.11ad concepts; CWNP 802.11 alternate PHYs white paper; IEEE Std 802.11

802.11ay: a wider, spatial 60 GHz system

802.11ad

one 2.16 GHz channel
directional single link

802.11ay

channel bonding + MIMO
 ≥ 20 Gb/s required mode

- Better beamforming, longer reach
- New jobs: fixed access, backhaul

Q: Who buys a radio that cannot cross a wall?

A: Anyone replacing a cable: docking, VR tethers, and — at 11ay scale — fixed wireless access and street-level backhaul.

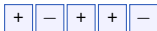
60 GHz evolved from cable replacement toward high-capacity access and backhaul.

beamforming = shaping the beam with antenna arrays (training finds the direction; beamforming makes the beam) backhaul = the link that carries a site's traffic back into the network

[IEEE Std 802.11ay-2021 \(enhancements above 45 GHz\); IEEE GEPS smart-cities paper](#)

Four waveform ideas explain the PHY story

DSSS / CCK



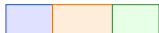
spread symbols across chips

OFDM



parallel orthogonal tones

OFDMA



assign tone groups to users

60 GHz SC



very wide directional carrier

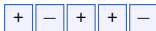
Sketches are conceptual, not calibrated time-domain or spectrum plots.

Waveform choice follows the channel: interference, delay spread, bandwidth, mobility, and hardware constraints.

IEEE Std 802.11; Cisco 802.11ax white paper; Keysight 802.11ad concepts

Four waveform ideas explain the PHY story

DSSS / CCK



spread symbols across chips

802.11b

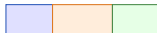
OFDM



parallel orthogonal tones

802.11a/g/n/ac

OFDMA



assign tone groups to users

802.11ax/be

60 GHz SC



very wide directional carrier

802.11ad/ay

Sketches are conceptual, not calibrated time-domain or spectrum plots.

Waveform choice follows the channel: interference, delay spread, bandwidth, mobility, and hardware constraints.

IEEE Std 802.11; Cisco 802.11ax white paper; Keysight 802.11ad concepts

The mainstream branch, b to be

802.11	year	GHz	MHz	key PHY idea	headline max	dimension spent
b	1999	2.4	≈22	CCK	11 Mb/s	spectrum
a	1999	5	20	OFDM	54 Mb/s	spectrum, const.
g	2003	2.4	20	OFDM	54 Mb/s	—
n	2009	2.4/5	20/40	OFDM + MIMO	600 Mb/s	space, BW
ac	2013	5	20–160	OFDM + MU-MIMO	6.9 Gb/s	BW, const., space
ax	2021	2.4/5/6*	20–160	OFDMA + MU-MIMO	9.6 Gb/s	scheduling
be	2024	2.4/5/6	20–320	EHT + MLO	≥30 Gb/s†	links, BW, const.

Years = amendment approval (802.11be published July 2025). * 6 GHz operation is in 802.11ax-2021; availability varies by national regulation (Wi-Fi 6E). spent = new dimension from the opening table (BW = bandwidth, const. = constellation).

† Required at the MAC data SAP — the honest counter: what survives preambles, ACKs, and retries. The ≈46 Gb/s PHY peak counts raw on-air bits.

Anchors, not promises — and the opening tally collected:
all six dimensions from slide one, spent.

IEEE 802.11 timeline; IEEE Std 802.11; IEEE Std 802.11be-2024; Cisco

A PHY rate is a product of optimism

$$\text{Rate} \propto \text{bandwidth} \times \text{bits/symbol} \times \text{code rate} \times \text{streams}$$

then subtract, from radio up to application:

**preamble +
pilots**

airtime overhead

**contention +
ACK**

MAC overhead

retries
channel loss

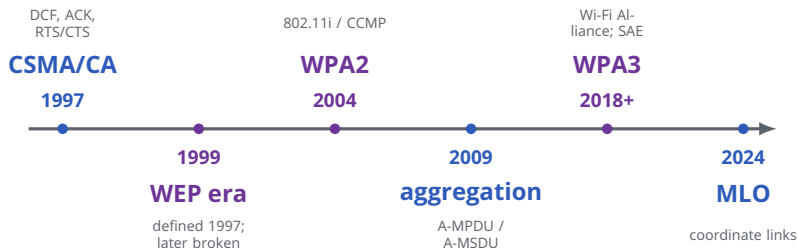
TCP / app
protocol overhead

Walking away from the AP, which factor on the top line fails first?

Application throughput is always lower — and depends on distance, interference, client count, and scheduler behavior.

Cisco, 802.11ac wireless throughput testing and validation

MAC and security evolved alongside



PHY innovation changes what is possible; MAC innovation decides who gets airtime — and how safely.

DCF = Distributed Coordination Function WEP/WPA = Wired Equivalent Privacy / Wi-Fi Protected Access CCMP = AES-based encryption of 802.11i SAE = Simultaneous Authentication of Equals

[IEEE Std 802.11; Wi-Fi Alliance security overview; IEEE Std 802.11be](#)

Wi-Fi and cellular optimize different edges

Wi-Fi

- Unlicensed spectrum
- You own the AP
- Contention-based access
- Very wide local channels
- Rapid device ecosystem

Cellular

- Licensed, managed spectrum
- Operator control
- Scheduled access
- Wide-area mobility
- Integrated service guarantees

Q: So which one wins?

A: Wrong question. Future edge systems choose, combine, and hand off across both — campus offload, AR/VR, industrial links, fixed wireless access, and edge computing already do.

IEEE GEPS, IoT and smart cities perspective

Read any generation in five questions

- 1 What spectrum does it use?
- 2 What does one PPDU look like?
- 3 How are users separated?
- 4 What channel conditions does it assume?
- 5 Which bottleneck moved next?

b → a/g → n → ac → ax → be (and ad → ay, off to the side)

Question 5's answers were the six rows of our opening table — each generation spent the next dimension.

Homework for your intuition: run the five questions on 802.11bn, the amendment being written right now.